

Exploring the Potential of Generative AI for Supporting Middle-Aged Individuals in Retirement Transitions

Minseo Park

Department of Industrial Design
KAIST
Daejeon, Republic of Korea
minseopark61@kaist.ac.kr

Youn-kyung Lim

Department of Industrial Design
KAIST
Daejeon, Republic of Korea
younlim@kaist.ac.kr

Abstract

The widespread adoption of Generative AI (GenAI) has fueled research exploring applications across diverse age groups and domains. However, its potential to support individuals in retirement transitions remains underexplored. This study aims to uncover the potential of GenAI in this context by examining its current and anticipated roles through semi-structured interviews with 10 middle-aged individuals navigating retirement transitions. The findings highlight three key roles of GenAI as an identity navigator, self-actualization facilitator, and connection catalyst. Building on the findings, the study identifies novel design opportunities for GenAI-based systems that assist an integrated journey from self-discovery to self-actualization, and support both direct and indirect connection to the world. The research aims to inspire the HCI community to further investigate these new design possibilities and attract attention to the unique context of retirement transition.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

generative AI, retirement transitions, middle-aged people, design opportunities

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1 Introduction

Generative Artificial Intelligence (GenAI) has become increasingly integrated into everyday life, with applications spanning content creation, business, education, and healthcare [12, 29]. Recent research has further explored its utility in personal reflection [14, 16] and self-care [6], exhibiting potential to aid middle-aged adults navigating significant life transitions [18, 34]. Recent initiatives such as workshops hosted by midlife and senior support centers nationwide

across South Korea and the United States demonstrate a growing interest in facilitating access to GenAI for this demographic [22]. However, there is a lack of empirical evidence on what roles GenAI can play in supporting transitional life phases for this group.

This study focuses on exploring the current roles of GenAI for middle-aged adults experiencing the transition to retirement and their expectations for future applications. Retirement transitions involve significant changes in social roles, personal values, and daily routines as individuals leave long-term employment and pursue new opportunities [8, 13], and has been recognized by ongoing research in HCI [11, 26, 27]. In contemporary contexts, these transitions also increasingly include midlife career shifts, where individuals retire from their primary occupation to pursue a second career, facilitated by extended life expectancy and greater career mobility [1, 3, 15, 32]. While the potential of GenAI to support career transitions among early-career individuals has been investigated [10], and its role in financial planning for retirement has been explored [21], its broader applications for addressing the unique needs of retirement transitions remain under-researched.

To address this gap, this study aims to answer the following research questions: 1) *What are the current and anticipated roles of GenAI in supporting retirement transitions?* 2) *What new design opportunities exist for GenAI systems in this context?* Semi-structured interviews with 10 middle-aged adults experiencing retirement transitions who are users of GenAI were conducted to investigate the two RQs.

Our findings identify three key roles of GenAI for supporting retirement transitions: GenAI as an identity navigator, a self-actualization facilitator, and a connection catalyst. These findings reveal significant opportunities for designing GenAI systems that offer integrated support from self-discovery to self-actualization, and provide both direct and indirect methods for connection. This study contributes to the HCI community by advancing the understanding of GenAI's potential in supporting individuals in retirement transitions, actively motivating further research on this topic.

2 Related Works

2.1 Retirement transition in HCI

The transition to retirement is widely recognized as a significant life transition for middle-aged and older adults, and research in HCI has increasingly focused on supporting this experience through technology [11, 26, 27]. Durrant's qualitative study highlights that internet use aids retirees in navigating transitions, maintaining relationships, and preserving autonomy [11]. Salovaara found that technologies like mobile phones and the internet help resolve conflicts between leisure and social priorities during retirement transitions

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[27]. Further, Qiu introduced EncoreLife, a gamified app addressing loneliness and loss of structure in retirees through collaborative activities and engaging features [26]. While HCI research shows the potential of technology to address challenges in retirement, there remains a lack of research on how technology can support retirees in discovering new roles and occupations after stepping away from primary careers. Recently, GenAI has emerged as a promising tool to provide individualized support, however research focusing specifically on its application in the retirement context remains scarce. This study investigates how GenAI is used and expected to support retirement transitions through accounts of users, aiming to identify its potential roles and expand its application for this critical life phase.

2.2 Generative AI for supporting life transition

While not specifically targeting middle-aged individuals, prior studies exploring the potential of GenAI in various contexts demonstrate its ability to support middle-aged adults navigating retirement transitions. These studies highlight GenAI's capacity to enhance well-being through personalized emotional support [6, 23, 28], facilitate self-reflection to help individuals reframe identity crises from new perspectives [2, 16, 17], generate simulations of future scenarios to enable indirect experiences of untried life paths [6, 10], and spark intimate conversations to foster social connections [7]. Du explored how GenAI-driven gamification could assist individuals during career transitions by enabling users to simulate hypothetical career scenarios, fostering enhanced self-awareness and planning [10]. Regarding retirement transitions specifically, prior research has explored GenAI's potential and challenges in supporting financial planning for the pre-retirement phase [21, 25]. However, less attention has been given to how GenAI could address broader aspects of retirement transitions, such as supporting identity reflection, facilitating the discovery and execution of self-fulfilling roles and activities, and maintaining connections with the world. This study aims to address this research gap by investigating the current accounts and expectations of middle-aged individuals utilizing GenAI. By grounding its findings in prior research, this study identifies the potential of GenAI-based systems to support retirement transitions for middle-aged adults, offering a meaningful contribution to this underexplored area.

3 Method

3.1 Semi-structured Interview

To investigate accounts and expectations regarding the use of GenAI to support retirement transition, we conducted semi-structured one-on-one interviews with 10 participants. Before the interviews, participants completed an online form designed to collect demographic information and their current uses of GenAI. The form specifically asked about how participants currently utilize GenAI and how it has been particularly helpful for retirement transition, encouraging them to reflect on their experiences before the interview. The interviews were conducted for an average of 61 minutes, using a mix of online and in-person formats. The interview questions were structured into three main sections: 1) participants' current experiences with retirement transitions, 2) their experiences using GenAI, and 3) their expectations for the role of GenAI in addressing their needs.

To prevent participants' varying AI literacy from limiting their thoughts on how GenAI can help them, participants were encouraged to imagine potential roles for GenAI without limiting their thoughts to its current capabilities. Informed consent was obtained from all participants for data use and study participation. Each participant received a compensation of 30,000 KRW. Consent for audio recording was sought before the interviews, and for one participant who declined to be recorded, detailed notes were taken manually instead. This study was approved by the Institutional Review Board (IRB) of the researchers' organization (IRB No.KAISTIRB-2024-21).

3.2 Participants

Participants were recruited based on two criteria: being middle-aged adults experiencing retirement transition and using GenAI. Researchers contacted and introduced the study to a local middle-aged support center that offers GenAI courses and requested assistance in recruitment. The study was also promoted through word of mouth. As a result, individuals who showed interest in sharing their experiences of retirement transition and using GenAI submitted the recruitment form (via Notion). Ultimately, 10 participants were recruited, with an average age of 51.1 years ($SD = 5.7$, $MIN = 41$, $MAX = 60$), comprising 4 males and 6 females. All participants were South Korean. To ensure diversity, the participants represented varying levels of GenAI usage and diverse retirement contexts (see Table 1). Participants used GenAI daily (P1, P3, P4), at least once a week (P2, P5, P8, P9), or at least once a month (P6, P7, P10). Each participant had a unique retirement context, with some having multiple reasons for their decision to retire. Reasons included unexpected termination (P1, P8), mandatory retirement due to age (P7, P10), perceived limitations caused by advancements in AI (P2, P4), caregiving responsibilities (P3, P9), relocation (P3), bereavement of a child (P6), physical limitations (P3, P5, P9), and lack of job fit (P4). At the time of the interviews, P1, P3, P5, and P6 were not employed after retirement but were seeking new career opportunities. P4 and P8 were enrolled in or planning to attend university courses to acquire new skills post-retirement. P2 and P10 were continuing their primary jobs while preparing for retirement, whereas P7, P8, and P9 had transitioned to new part-time roles after retirement.

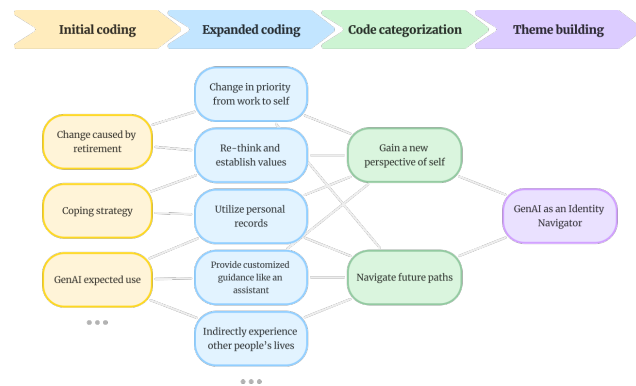
3.3 Data Analysis

After completing all interviews, audio recordings were transcribed using the Naver Clova Voice-to-Text service, ensuring that all participant information was anonymized. To qualitatively analyze the transcripts through thematic analysis [5], the first author conducted coding in multiple stages (see Figure 1). To gain a comprehensive understanding of the data, a memoing process was first conducted, followed by initial coding to categorize the data. For example, initial codes such as 'Change caused by retirement,' 'Coping strategy,' 'GenAI current use,' and 'GenAI expected use' were created. Based on the initial codes, expanded codes were developed to capture the deeper meaning of the quotes, such as 'Change in priority from work to self,' 'Find new social connections,' 'Gain answer to philosophical questions,' and 'Indirectly experience other people's lives.' Subsequently, the final codes were grouped, with all researchers engaging in iterative and in-depth discussions for the final themes. To

Table 1: Participant information

ID	Gender	Age	Retirement context	GenAI usage frequency	GenAI used
P1	M	54	Retired from working as a caregiver, searching for a new job	Everyday	ChatGPT, Perplexity, Copilot
P2	M	49	Currently working as a screenwriter/artist, preparing to retire	More than once a week	ChatGPT, Suno
P3	F	41	Retired from working as a digital marketer, searching for a new job	Everyday	ChatGPT, Gemini
P4	F	46	Retired from working as an IT consultant, now studying AI creator course	Everyday	ChatGPT, Claude, Copilot
P5	F	53	Retired from working as an English teacher	More than once a week	ChatGPT
P6	F	55	Retired from administrative work, searching for a new job	More than once a month	ChatGPT
P7	M	60	Retired from primary job (military service for 33 years), re-employed as facilities management staff	More than once a month	ChatGPT
P8	F	45	Retired from primary job (bank clerk), re-employed as a storyteller for children	More than once a week	ChatGPT
P9	F	53	Retired from primary job (office work), re-employed as a technology mentor for older adults	More than once a week	ChatGPT, Gemini, Copilot
P10	M	55	Currently working in the broadcasting industry (for 31 years), preparing to retire	More than once a month	ChatGPT

identify the roles of GenAI in supporting retirement transition experiences, codes related to understanding retirement transition experiences were first grouped. Then, codes related to GenAI use that intersected with retirement transition experiences were selected and grouped. Grouping these categories, three themes representing the core roles of GenAI for supporting retirement transitions and seven sub-themes to further specify the roles were constructed. Then, representative quotes were extracted for each theme. Finally, all quotes and theme names were carefully translated into English when organizing the research findings, with two researchers conducting cross-check reviews to ensure accuracy and clarity.

**Figure 1: Coding and theme building procedures for theme ‘GenAI as an Identity Navigator’ (selected examples)**

4 Findings

The potential of Generative AI in supporting participants through retirement transitions was identified across domains including self-reflection, new career exploration and execution, lifelong learning, social contributions, and maintaining societal connections. Within these domains, three key roles of GenAI emerged: 1) **Identity Navigator**, aiding in redefining values and setting future directions; 2) **Self-Actualization Facilitator**, reducing barriers to unfamiliar activities; and 3) **Connection Catalyst**, enabling connections to both local and unfamiliar relationships and resources. Each role is detailed in the following subsections.

4.1 GenAI as an Identity Navigator

As participants transitioned into or prepared for retirement, they commonly experienced a significant shift in values and identity, moving from a life centered on external factors such as work, family, and organizational life to one centered on inner purpose. Generative AI played (or was expected to play) a supportive role in coping with this change as an **identity navigator**, **aiding participants in understanding themselves from new perspectives, clarifying personal values, and setting directions for their future.**

4.1.1 Gain a new perspective of self. Participants viewed GenAI as a tool that could analyze and understand their personal information, helping them gain new insights into themselves. P4 used ChatGPT’s custom GPTs feature [24] to train a personalized GenAI agent with over 100 of her own essays she has been writing for over 10 years. Through conversations with this customized agent, P4 asked questions about her own behavior, personality, and values, gaining fresh perspectives on GenAI’s interpretation. Additionally, P4 reflected on how interactions with the agent revealed and corrected long-held beliefs or values that had gone unexamined

over time. *“While negative experiences from childhood solidify into negative beliefs, they’re not necessarily valid in the present. GenAI offered me new perspectives, which helped correct my thinking”* (P4). P2 asked ChatGPT philosophical questions while contemplating the second chapter of life. He expressed surprise at its ability to answer challenging questions like *“What does it mean to live a truly happy life?”* and found that its responses provided opportunities to rethink and solidify his own values. This shows that GenAI has potential to enable meaningful conversations by generating responses to complex questions tailored to one’s own interest, and re-thinking one’s identity through GenAI’s analysis of personal data.

4.1.2 Navigate future paths. Participants expressed future expectations for GenAI to function as a personal counselor, leveraging their personal data to uncover one’s untapped talents (P5). Additionally, they hoped GenAI could learn about their philosophy and personal interests through long-term conversations starting before retirement and use this understanding to analytically recommend new careers, hobbies, or activities to help design a second chapter in life (P9, P10). *“If someone continuously uses an AI assistant while working, the AI would naturally come to understand what that person is interested in and what they are curious about. In a way, the AI would have deep knowledge of the person’s philosophy and interests. I think it could then provide more scientific and analytical options to help design the second phase of life.”* (P10) These insights reveal participants’ expectations and needs for personalized recommendations based on their personal data to help establish new life directions.

4.2 GenAI as a Self-Actualization Facilitator

For the participants, retirement marked both the end of one chapter of life and the beginning of another. In this newly embraced second act, they pursued new careers, hobbies, education, and social contribution activities they had genuinely desired to do for a long time. This process was often preceded by a stage of self-discovery. *“I’m in my 50s, and I figure I have about 10 more years to work. If things go well, maybe I could stretch that a bit longer. But with the time I have left in life, I’ve been thinking more about what kind of work I can do that truly feels fulfilling. That’s why I think I’m now focusing on something that’s closer to true self-actualization.”* (P5) However, venturing into unfamiliar fields posed significant challenges, and GenAI served (or was expected to serve) as a **self-actualization facilitator that helped them pursue new opportunities by lowering the barriers of unfamiliar activities.**

4.2.1 Facilitate new career execution. Participants discussed the psychological barriers associated with attempting completely unfamiliar careers that one has desired internally, and expressed interest in using GenAI to lower these barriers by simulating job experiences. P10 envisioned a GenAI-based system that could provide customized indirect experiences: *“It can provide insights from the perspective of someone in that role. It could help people experience what it’s like to handle the responsibilities of a job in advance.”* (P10) Currently, P3, P6, and P9 utilized GenAI to assist with writing resumes and cover letters for reemployment. They input their career histories and sought guidance on tailoring their experiences to align with the concept of the target company (P9) or asked for help

in highlighting their personal background and achievements (P3, P6). This demonstrates GenAI’s potential in helping individuals effectively *“personal-brand”* (P3) based on their past experiences and translate it into compelling written narratives. Participants also used GenAI to develop new skills for reemployment or to complement their lacking abilities. P2, who desired to explore careers in video production after retirement, shared excitement about using GenAI (e.g., Suno [31]) to create music, which has been his long-held aspiration. *“When I was 20, I wanted to try composing music and even bought a book about it, but I gave up. Now, at almost 50, I’m finally making music, and it’s such a joy... I can create my own music with the help of AI.”* (P2) These highlight the role of GenAI in empowering individuals to overcome the difficulties of pursuing new career opportunities and skill gaps.

4.2.2 Facilitate lifelong learning. Participants actively used GenAI to do hobbies and learn new things, such as traveling, exercising, and studying languages, enabling them to take on challenges they previously found difficult to pursue on their own. For instance, P7 used GenAI to learn about the country he is planning to visit, while P5 sought advice on workout routines and followed it as suggested. P4 expressed that GenAI opened up limitless potential to take on new challenges whenever they desired: *“If there’s something I want to do, it tells me exactly how to approach it. I now have so many things I want to learn, even in completely new areas. In the past, I wouldn’t have known how to start due to lack of information, but now, as long as I have the time and energy, I can do anything I’m curious about.”* (P4)

4.2.3 Facilitate social contribution. The majority of participants expressed a desire to engage in activities that contribute to society during their second phase of life after retirement (P2, P3, P5, P8, P9, P10). Now retired, they viewed this as the ideal time to fulfill their aspirations for volunteering. Some participants explored ways to leverage their accumulated strengths to make meaningful contributions (P5, P10), but encountered challenges in identifying and executing suitable opportunities. P10 expressed enthusiasm to write a book aimed at sharing his professional knowledge and experiences to the society, which GenAI can help by *“turning fragmented information to logical structured writing”* (P10), and provide structural guidance and context personalized questions that *“help me understand my inner thoughts”* (P10). This suggests the potential of GenAI acting as a facilitator for organizing retiree’s own experiential legacy, offering a way to contribute to the society.

4.3 GenAI as a Connection Catalyst

For retired middle-aged adults, maintaining social connections is another critical area where support is needed. Participants reported significant changes in their existing social relationships and information network, expressing fears of isolation and disconnection. To address this, GenAI played (or was expected to play) the role of a **connection catalyst by fostering connections to local communities and information outside one’s usual boundaries.**

4.3.1 Connect to local communities. Participants sought support for staying connected to the world after retirement by attending local community centers, where they met new people and enrolled in educational programs (P1, P2, P3, P5, P6, P7, P8, P9). P7, for instance,

used GenAI to connect with local networks offering the desired educational programs. *"I asked (ChatGPT) about places nearby where I could learn Japanese outside of traditional language schools, specifying my conditions. That's how I found a local cultural center. Without this, I would've had to rely on word-of-mouth information through people I know."* (P7) The ability to tailor searches to specific conditions highlighted GenAI's advantage in improving accessibility to tailored information compared to general search engines. However, P7 also raised concerns about the reliability of GenAI when seeking highly localized information: *"The responses sometimes change slightly with each query, and occasionally the information isn't accurate. It reduced trustworthiness."* (P7) This illustrates the challenges posed by GenAI's current limitations, which can hinder its utility for connecting users to local networks and resources.

4.3.2 Connect to the world outside my boundaries. Participants expressed a desire for GenAI to play a role not only in connecting them to their local communities but also in keeping them engaged with global trends and a more diverse range of people. P10, for instance, had previously worked alongside individuals from various age groups, naturally staying informed about the latest trends. However, he voiced concerns about losing this connection after retirement and expressed hopes that GenAI could actively expose him to real-time trends. P3 emphasized the need for tools and education to challenge fixed mindsets and adapt to a rapidly changing world. She highlighted GenAI's potential to serve as an accessible means to facilitate this adaptation: *"The world and technology are changing so quickly now. Many of us have become set in our ways after building expertise in one field, so we need tools that can make our thinking more flexible and open to adopting new thoughts and technologies."* (P3) To achieve this, P3 also proposed fostering a culture of mutual respect and communication across generations and expressed anticipation for GenAI to help with such connections.

While participants generally expressed satisfaction with using GenAI to connect to the world, they also voiced concerns about over-reliance. P3 and P10 emphasized the irreplaceable value of direct human interaction and solidarity: *"Spending an hour searching with ChatGPT can provide a lot of information, but often, meeting an expert in the field over a cup of tea can yield far more valuable insights and advice."* (P10) These insights suggest that while GenAI offers significant potential as a tool for increasing accessibility for information, there is a need to balance its use with opportunities for direct human interaction to avoid over-dependence and preserve the essence of human relationships.

5 Discussion

Our findings reveal the potential of GenAI to support middle-aged individuals in retirement transitions, especially in three key roles as an identity navigator, self-actualization facilitator, and connection catalyst. Based on these findings and previous research, we propose two design opportunities of GenAI-based systems to empower individuals navigating this life stage.

5.1 Support Self-Discovery to Self-Actualization

Participants anticipated GenAI's role in helping them reflect on their changing identities and view themselves from new perspectives, ultimately envisioning GenAI to guide future life directions

(Section 4.1). Through self-reflection and personalized path recommendations, participants could identify the life paths they genuinely desire and then use GenAI to support their actualization efforts (Section 4.2). As these two processes both hold importance and are complementary (Section 4.2), future work could explore integrated systems that enable seamless support for the entire journey from self-discovery to self-actualization. The potential of GenAI in providing such support is evidenced by prior studies. GenAI can interpret accumulated personal data to generate content offering fresh perspectives, pose contextually relevant questions during conversations to foster deep self-reflection, rather than merely offering fragmented question-and-answer exchanges [2, 6, 14, 16]. Additionally, utilizing this introspection data, GenAI can recommend tailored pathways for a new phase of life, and provide personalized simulations of indirect experiences to aid decision making [10]. Lastly, GenAI can further support actualization of new opportunities by helping users overcome the initial barriers by complementing lacking abilities and providing frameworks [30]. By leveraging the rich life experiences of middle-aged individuals, GenAI can provide highly personalized support throughout the flow from introspection to facilitation. We encourage future research to build on this opportunity, focusing on developing design guidelines and addressing challenges for creating GenAI systems that empower individuals in retirement transitions with this integrated process.

5.2 Support Direct and Indirect Connection

Participants highlighted the need for GenAI to address changes in their social networks by facilitating both direct connections to local communities and indirect connections by providing access to world wide information (Section 4.3). While direct connections are essential for fostering social bonds and accessing local resources, they may be limited in expanding perspectives beyond personal boundaries. Conversely, over-dependence on GenAI as an information source may lead to undermining the value and solidarity provided by direct human interactions (Section 4.3.2). Therefore, a balanced integration of these connection types is essential, with GenAI serving not only as an information provider but also as a facilitator of human-to-human connections. GenAI systems could facilitate direct connection by integrating with local services (e.g., map or community platforms) to provide accurate and personalized connection opportunities to local communities [33], which can mitigate the reliability issues discussed in Section 4.3.1. Currently, GenAI could actively generate curated daily summaries of trending information [19, 20] to facilitate connection to wider sources. Furthermore, GenAI could enable intergenerational communication as a source for connection outside ones' boundaries, as suggested by one of our participants (Section 4.3.2). GenAI can aid this connection by enhancing mutual understanding through adapting conversation styles to each other, providing context relevant timely prompts to facilitate dialogue, and offering interventions to manage potential conflicts [7, 9]. Leveraging GenAI's unpredictable and creative qualities [30], GenAI based systems could also promote a more indirect connection by implicitly revealing the interests or viewpoints of strangers, providing new perspectives and broadening understanding of the world [4]. We encourage future exploration of GenAI systems that harmonize these direct and

indirect connection opportunities for individuals transitioning to retirement.

6 Limitations and Future Work

Our study has several limitations. The study focused on middle-aged individuals in South Korea, which may introduce cultural bias. Additionally, six of the ten participants were recruited from a single local support center, potentially representing a group with higher community accessibility and a shared interest in post-retirement careers. The relatively young mean age of participants (51.1 years) also limits the generalizability to older individuals experiencing retirement transitions. Individuals unfamiliar with GenAI may face additional barriers such as limited accessibility to technological resources, which were not addressed here. Future research should include participants from diverse cultural backgrounds, older age groups, and non-user groups for a more comprehensive understanding.

In this study, we adopted a broad definition of retirement, including cases where middle-aged individuals leave their primary job to pursue a new career. As a result, our study encompasses a diverse range of retirement contexts. Given that retirement experiences and the role of GenAI may vary depending on specific circumstances (e.g. retiring due to reaching official retirement age or retiring to pursue a new career path), we suggest future research focus on a more specific group to explore the context in greater depth. Furthermore, our participants predominantly used text-based GenAI tools, and usage of multi-modal generation such as image and sound was limited and secondary. Further exploration into the potential of multimodal GenAI in this context is recommended.

This study serves as a preliminary exploration aimed at understanding the potential of GenAI for supporting retirement transitions. Due to this goal and the voluntary nature of the study, recruited participants were generally experiencing the positive side of its use more than possible challenges. Additionally, participants relied on memory to describe their GenAI usage, which lacked revealing in-the-field engagements with GenAI. As a next step, we plan to build on the potential identified by applying in-the-field investigation through methods like diary study with technology probes, through which we hope to identify design considerations that encompass both the possibilities and challenges of designing GenAI-based systems for this context.

7 Conclusion

This study explores the potential of GenAI to support middle-aged individuals transitioning into retirement by investigating its current and anticipated roles. Through semi-structured interviews with 10 middle-aged individuals navigating retirement transitions and using GenAI, the research highlights three key roles of GenAI in this context: identity navigator, self-actualization facilitator, and connection catalyst. These findings reveal opportunities for designing GenAI-based systems that assist self-discovery to self-actualization, and support both direct and indirect connection to the world. This study contributes to the understanding of GenAI's role in this critical life phase and seeks to inspire the HCI community to further investigate the design opportunities to support individuals in retirement transitions.

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